

CSE_2323

Sequential Logic Flip Flop

Digital Logic Design

Course code: CSE-2323

Credit Hour: 3

Md. Mujibur Rahman Maruf

Assistant Lecturer,

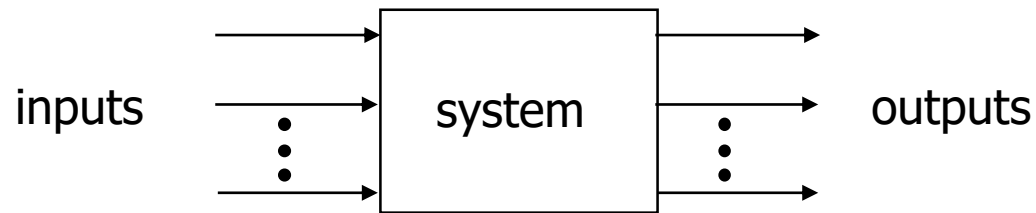
Dept. of CSE, IIUC

Mujiburmaruf.cuet17@gmail.com

International Islamic University Chittagong(IIUC)

Combinational vs. sequential systems

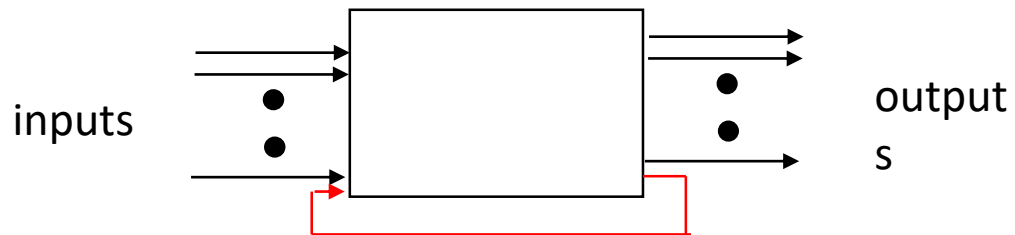
A simple model of a digital system is a unit with inputs and outputs:



Combinational systems are "memory-less" The outputs depend only on the present inputs

Sequential systems have memory

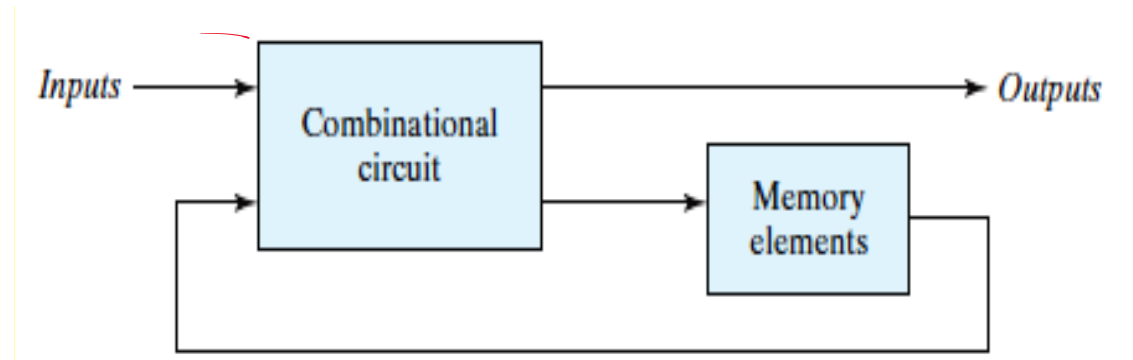
The output values depend on the input values and previous input values



Sequential Logic

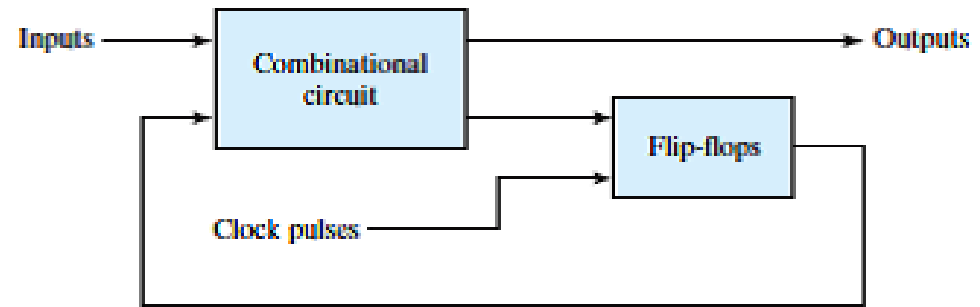
There are **TWO** main types of sequential circuits:

➤ **Asynchronous sequential circuit:** Sequential circuits that don't use a clock signal to determine the timing of their operations



Synchronous sequential circuit:

outputs and internal states are changed synchronously following a clock signal.

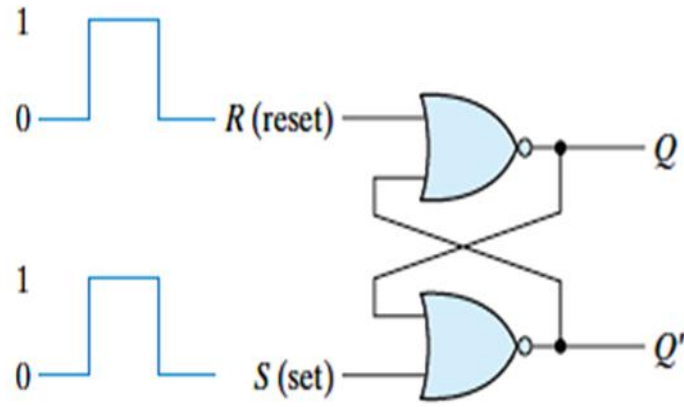


(a) Block diagram



(b) Timing diagram of clock pulses

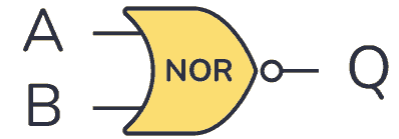
S R Latch: NOR Gate



(a) Logic diagram

S	R	Q	Q'
1	0	1	0
0	0	1	0
0	1	0	1
0	0	0	1
0	0	1	0
1	1	0	0

(b) Function table



A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

NOR gate truth table

Case:1

- When $S=0$ and $R=1$, the latch is reset into a state where $Q=0$ and $Q'=1$
- When $R=S=0$ the circuit remains in its current state ($Q=1$ and $Q'=0$)

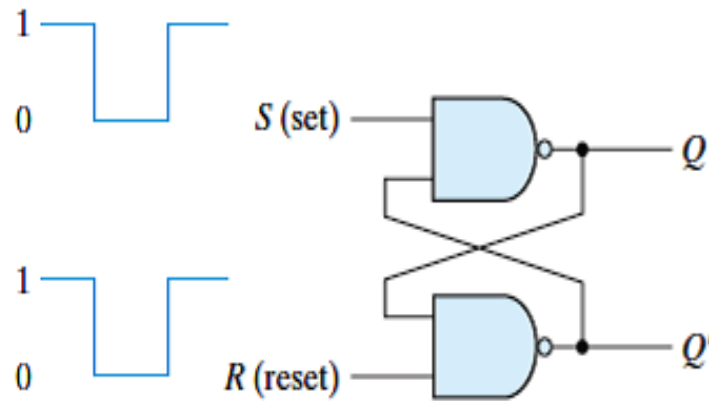
Case:2

- When $S=1$ and $R=0$, the latch is set into a state where $Q=1$ and $Q'=0$
- When $R=S=0$ the circuit remains in its current state ($Q=1$ and $Q'=0$)

Case 3:

Where $S=1$ and $R=1$, $Q_2=Q=0$ (there are actually problems with this state as we will see)

S R Latch: NAND Gate



(a) Logic diagram

S	R	Q	Q'
1	0	0	1
1	1	0	1 (after $S = 1, R = 0$)
0	1	1	0
1	1	1	0 (after $S = 0, R = 1$)
0	0	1	1 (forbidden)

(b) Function table



A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

NOR gate truth table

Case:1

- When $S=0$ and $R=1$, the latch is reset into a state where $Q=1$ and $Q'=0$
- When $R=S=1$ the circuit remains in its current state ($Q=1$ and $Q'=0$)

Case:2

- When $S=1$ and $R=0$, the latch is set into a state where $Q=0$ and $Q'=1$
- When $R=S=0$ the circuit remains in its current state ($Q=0$ and $Q'=1$)

Case 3:

Where $S=0$ and $R=0$, $Q_2=Q=1$ (there are actually problems with this state as we will see)

Flip flop

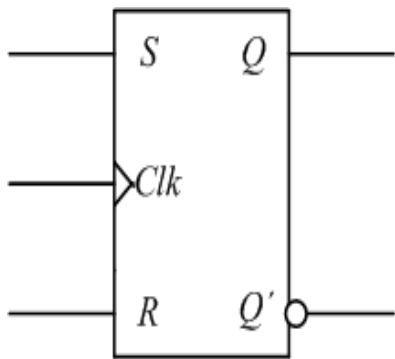
A device that stores either a 0 or 1.

Flip-flop :is a clocked device, in which only the clock edge determines when a new bit is entered.

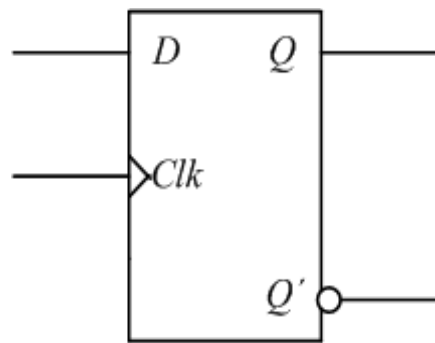
Stored value can be changed only at certain times determined by a clock input.

New value depend on the current state and it's control inputs

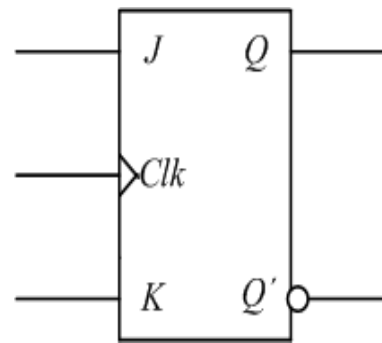
A digital circuit that contains flip-flops is called a **sequential circuit**



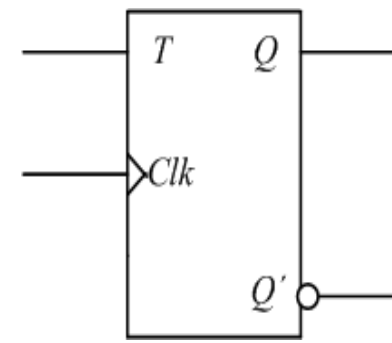
S-R flip-flops



D flip-flop

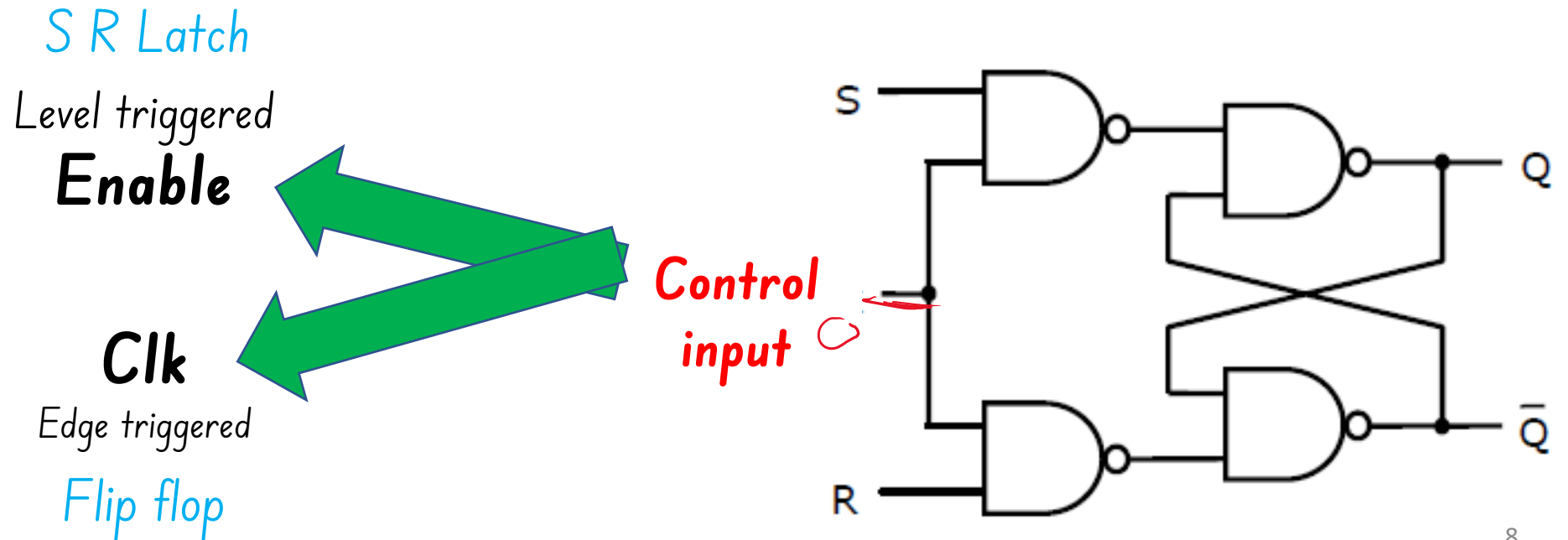


J-K flip-flops



T flip-flop

Difference between Latch and Flip Flop

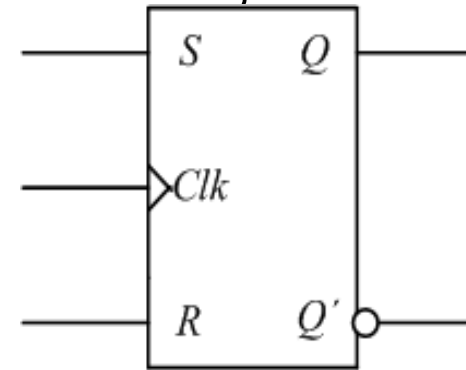
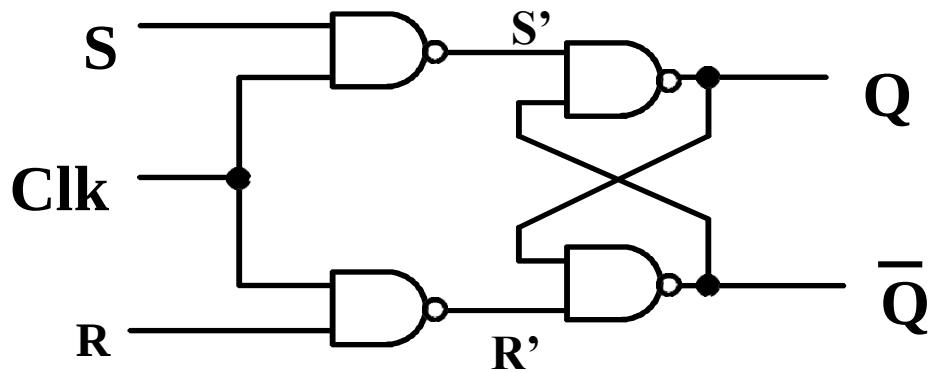


Difference between Latch and Flip Flop

Feature	Latch	Flip-Flop
Triggering	Level-triggered (responds to input as long as enable is active)	Edge-triggered (responds only at the rising/falling edge of a clock pulse)
Control Signal	Controlled by Enable signal	Controlled by a Clock signal
Timing Sensitivity	Continuous input monitoring	Discrete input sampling
Speed	Generally faster (less circuitry)	Slower due to clock dependency
Design Complexity	Simpler	More complex
Usage	Used where speed and simplicity matter	Used in synchronous systems (e.g., CPUs)

S - R Flip Flop

- Adding two NAND gates to the basic S - R NAND latch gives the clocked S'-R' Latch



S-R flip-flops

Truth Table			
clk	S	R	Q_{t+1}
0	x	x	Q_t (Memory)
1	0	0	Q_t (Memory)
1	0	1	0
1	1	0	1
1	1	1	Invalid

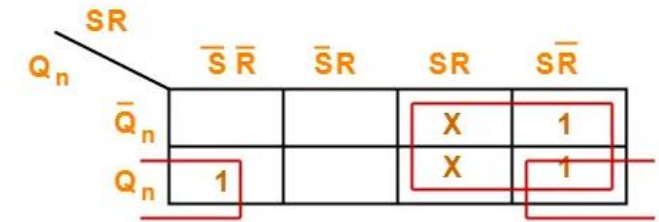


- $S = 0, R = 0 \rightarrow$ No change: The output **retains** its previous state. This is also called the **memory state**.
- $S = 0, R = 1 \rightarrow$ Reset state: Output $Q = 0$, and $Q' = 1$.
- $S = 1, R = 0 \rightarrow$ Set state: Output $Q = 1$, and $Q' = 0$.
- $S = 1, R = 1 \rightarrow$ Invalid condition: Both outputs go to the **same value**, which breaks the basic rule that **Q and Q' must be complements** of each other. This is considered an **unstable or undefined state**.

S - R Flip Flop

A **characteristics table** shows the next state of a flip-flop (Q_{n+1}) based on its current state (Q_n) and the input(s).

An **excitation table** shows the required **input(s)** to achieve a desired **transition** from the current state (Q_n) to the next state (Q_{n+1}).

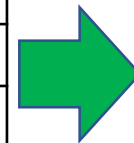


K Map $Q_{t+1} = S + Q_t R'$

Truth Table			
clk	S	R	Q_{t+1}
0	x	x	Q_t (Memory)
1	0	0	Q_t (Memory)
1	0	1	0
1	1	0	1
1	1	1	Invalid



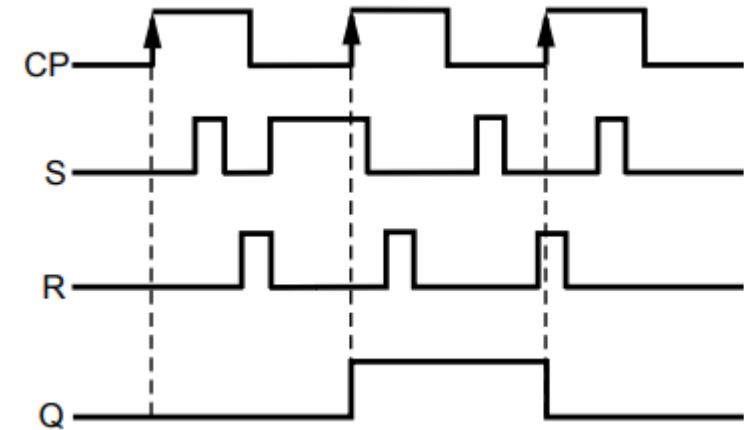
characteristics Table				
Q(t)	S	R	Q(t+1)	Comment
0	0	0	0	No change
0	0	1	0	Clear Q
0	1	0	1	Set Q
0	1	1	X	Indeterminate
1	0	0	1	No change
1	0	1	0	Clear Q
1	1	0	1	Set Q
1	1	1	x	Indeterminate



SR Flip-flop			
Q(t)	Q(t+1)	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

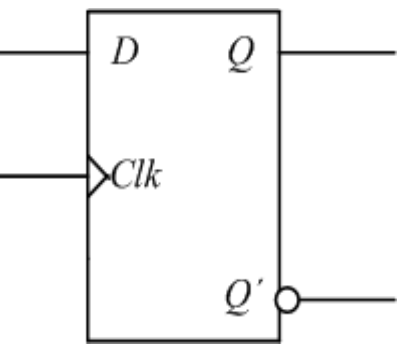
Example- S - R Flip Flop

- The waveform diagram illustrates the behavior of a **positive edge-triggered** clocked SR flip-flop.
- The clock pulse (CP) has rising edges marked with arrows where the flip-flop evaluates inputs.
- At each rising edge, the state of inputs S (Set) and R (Reset) determines the output Q.**
- On the first rising edge, $S = 0$ and $R = 0$, so Q is reset to 0.
- On the second rising edge, $S = 1$ and $R = 0$, so Q is set to 1.
- On the Third rising edge, $S = 0$ and $R = 1$ again, so Q is reset to 0.
- The output Q changes only at the clock's rising edges and ignores S and R changes in between.
- This demonstrates the edge-triggered operation of the SR flip-flop, responding only at clock transitions.

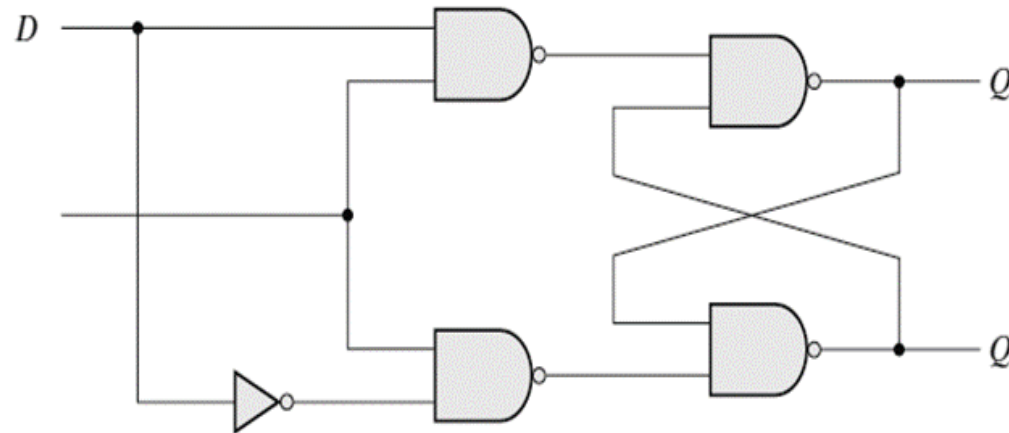


D Flip Flop

The D stands for "data"; this flip-flop stores the value that is on the data line.



Clk



(a) Logic diagram

Clk	D	Next state of Q
0	X	No change
1	0	$Q = 0$; reset state
1	1	$Q = 1$; set state

(b) Function table

D flip-flop

characteristics Table

Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

Q_n	D	
	0	1
0		1
1		1

$$Q_{n+1} = D$$

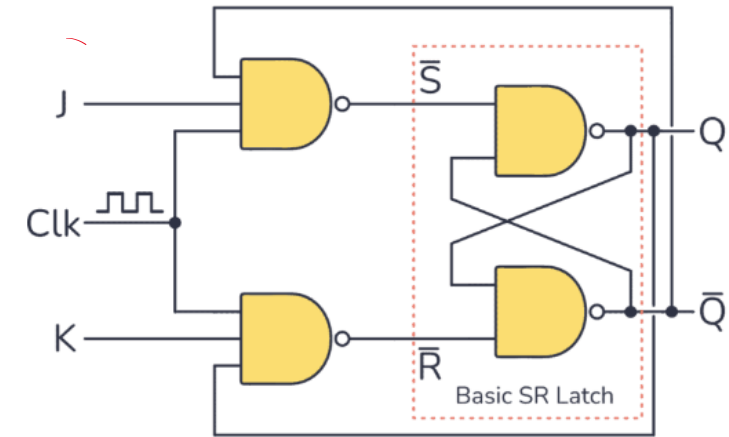
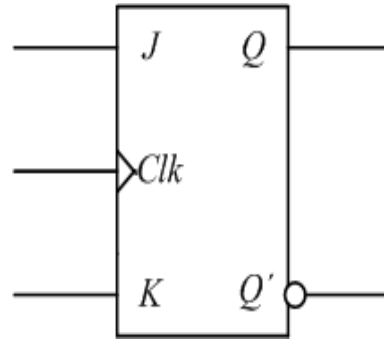
Excitation Table

D Flip-flop		
Q(t)	Q(t+1)	DR
0	0	0
0	1	1
1	0	0
1	1	1

J K Flip Flop

J and K inputs of the JK flip-flop can be used to set, reset, or toggle the output, like this:

- $J=1$ and $K=0$ sets the output to 1
- $J=0$ and $K=1$ reset the output to 0
- $J=1$ and $K=1$ toggle the output



JK Flip-Flop basic circuit

J-K flip-flops

JK Flip Flop Truth Table

Truth Table			
clk	J	K	Q_{t+1}
0	x	x	Q_t (Memory)
1	0	0	Q_t (Memory)
1	0	1	0
1	1	0	1
1	1	1	Q'

- For CLK=1 (Flip Flop is Enabled)
- For $J=0, K=0 \Rightarrow \bar{S}=1, \bar{R}=1 \Rightarrow$ Latch Retains Values.
- For $J=0, K=1$
 - If $Q = 1, \bar{Q} = 0 \Rightarrow \bar{S}=1, \bar{R}=0 \Rightarrow$ Latch Resets.
 - If $Q = 0, \bar{Q} = 1 \Rightarrow \bar{S}=1, \bar{R}=1 \Rightarrow$ Latch Retains Reset Mode.
- For $J=1, K=0$
 - If $Q = 1, \bar{Q} = 0 \Rightarrow \bar{S}=1, \bar{R}=1 \Rightarrow$ Latch Retains Set Mode.
 - If $Q = 0, \bar{Q} = 1 \Rightarrow \bar{S}=0, \bar{R}=1 \Rightarrow$ Latch Sets.
- For $J=1, K=1$
 - If $Q = 1, \bar{Q} = 0 \Rightarrow \bar{S}=1, \bar{R}=0 \Rightarrow$ Set Mode Toggles.
 - If $Q = 0, \bar{Q} = 1 \Rightarrow \bar{S}=0, \bar{R}=1 \Rightarrow$ Reset Mode Toggles.

J K Flip Flop

characteristics Table				
Q(t)	J	K	Q(t+1)	Comment
0	0	0	0	No change
0	0	1	0	Clear Q
0	1	0	1	Set Q
✓0	1	1	1	toggle
1	0	0	1	No change
1	0	1	0	Clear Q
✓1	1	0	1	Set Q
1	1	1	0	toggle



Excitation Table

JK flip-flop			
Q(t)	Q(t+1)	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

For $Q_{(t+1)}$

	$\bar{J}\bar{K}$	$\bar{J}K$	JK	$J\bar{K}$
$\bar{Q}$			1	1
Q	1			1

$$Q_{t+1} = J\bar{Q} + \bar{K}Q$$

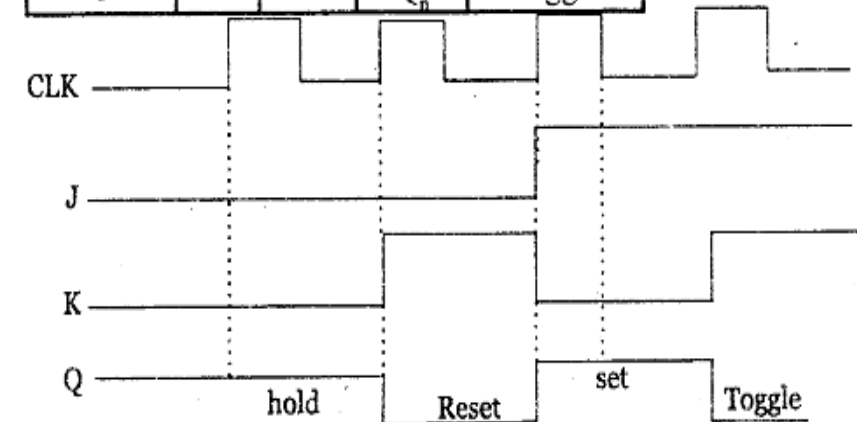
J K Flip Flop

The image shows the **timing diagram** and **truth table** for a **positive edge-triggered JK flip-flop**. The **truth table** defines how the JK flip-flop behaves based on the J, K, and clock (CLK) inputs. The flip-flop is **edge-triggered**, meaning it only responds to inputs on the **rising edge of the clock** ($\uparrow$).

1. The **first rising edge** of CLK: $J = 0, K = 0 \rightarrow$ Output **Q holds** its previous state.
2. The **second rising edge**: $J = 1, K = 0 \rightarrow$ According to the table, Q is **set to 1**.
3. The **third rising edge**: $J = 0, K = 1 \rightarrow$ Q is **reset to 0**.
4. The **fourth rising edge**: $J = 1, K = 1 \rightarrow$ The flip-flop **toggles** its output.

Truth table.

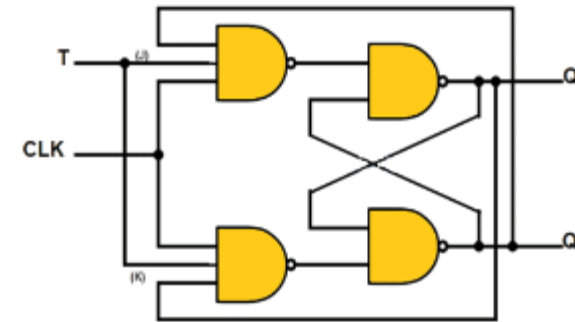
CLOCK	J	K	Q_{n+1}	Action
0	X	X	Q_n	No change
1	0	0	Q_n	No change
1	1	0	1	Reset
1	1	1	Q_n	Toggle



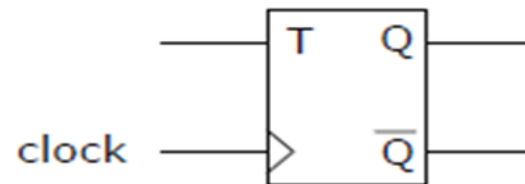
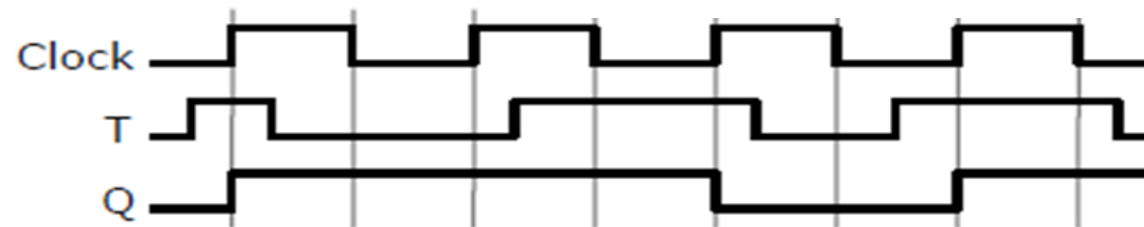
T Flip Flop

The name *T* derives from the behavior of the circuit, which 'toggles' its state when $T=1$

- This feature makes the *T* flip-flop a useful element when constructing counter circuits



Input	Present State	Next State
T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0



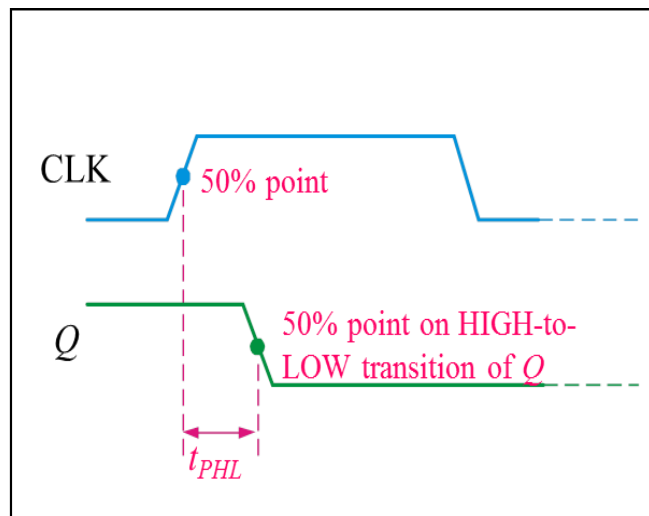
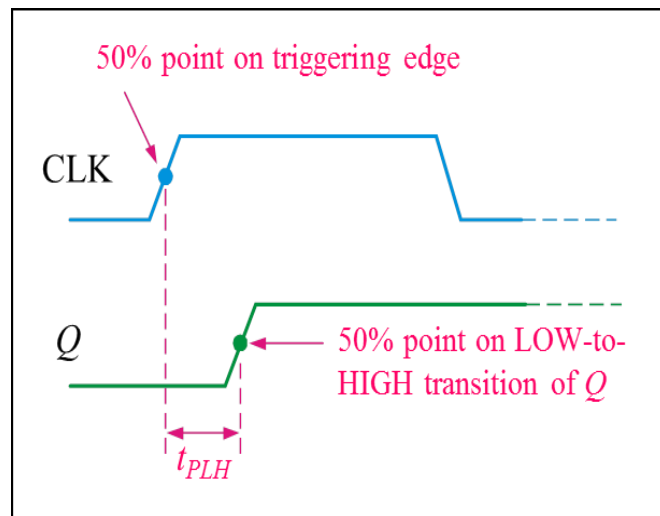
Positive edge triggered

$Q_n \backslash T$	0	1
0		1
1	1	

$$Q_{n+1} = T \cdot \overline{Q_n} + \overline{T} \cdot Q_n$$

Propagation Delay

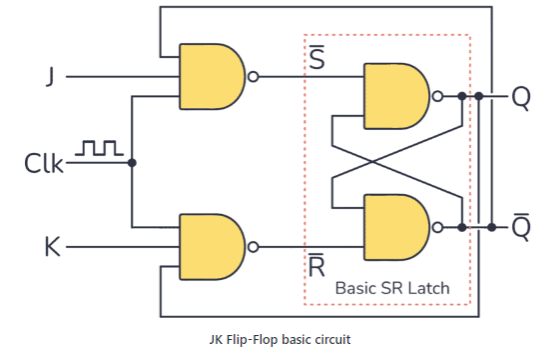
Propagation delay time is specified for the rising and falling outputs. It is measured between the 50% level of the clock to the 50% level of the output transition



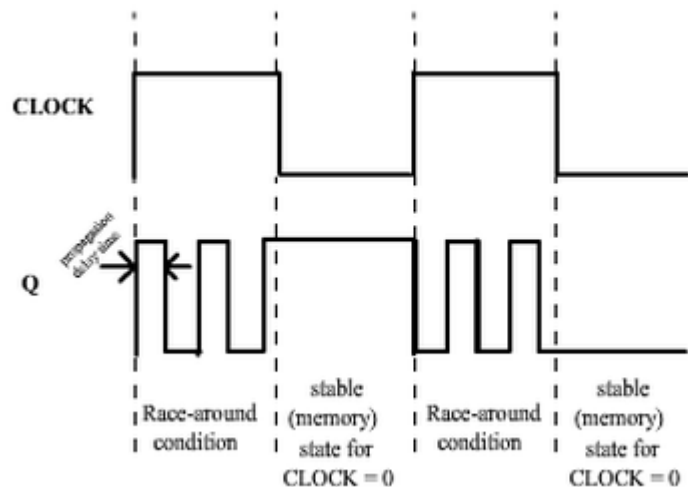
Race around condition

Truth table of JK flip flop was formed with the assumptions that inputs do not change during clock pulse.

- But this condition is not true because of feedback connections.
- Due to the feedback connection there is uncontrolled toggling at the output.



Race Around Condition In JK Flip-flop - For J-K flip-flop, if $J=K=1$, and if $clk=1$ for a long period of time, then Q output will toggle as long as CLK is high, which makes the output of the flip-flop unstable or uncertain. This problem is called race around condition in J-K flip-flop.



Solutions of Racing

1. Clock Pulse Duration $\neq$ Propagation Delay of NAND gates
(not feasible)
2. Edge triggered Flip flop
3. Master-Slave JK Flip flop

Race around condition

